

A Robust System for Thermoelectric Device Characterization

João Ider^{ID}, Adhimar Oliveira^{ID}, and Rero Marques Rubinger^{ID}

Abstract—Due to the large reduction in fossil fuel reservoirs, the consequent cost increase of deepwater extraction, and the emission of pollutants, there is a constant search for alternative ways to obtain clean energy at a lower cost. Among those sources, we focus on the energy produced by thermoelectric materials. In this work, we present a new system for the characterization of thermoelectric generation devices. Such a system performs measurements of electrical resistivity, Seebeck coefficient, and thermal conductivity in a single setup. With this, it is possible to reduce the systematic errors in the figure of merit ZT and the cost of the equipment. Our equipment, together with the developed software, presented excellent results and analyses, and with that, it proves to be a robust alternative for the characterization of commercial thermoelectric devices, and of laboratory thin film thermoelectric materials.

Index Terms—Electrical conductivity, figure of merit, Seebeck coefficient, systematic errors, thermal conductivity, thermoelectric device characterization.

I. INTRODUCTION

THE increasing search and need for new environment friendly energy sources drives research on thermoelectric materials and characterization techniques. Among the materials studied, thermoelectric power plants have emerged as a promising solution. The advantages are: being small, having no moving parts, and performing conversion of heat into electricity with a single module. Numerous researches have been dedicated to looking for high-performance and low-cost thermoelectric materials [1]. The search for such materials or to improve their ratings is continuously growing due to the strong global demand for economically viable and pollution-free energy conversion [2]–[5].

Energy is a commodity with worldwide increasing consumption and being able to harvest energy considering environmental issues, energy has attracted great attention in recent years. To overcome the energy crisis that is approaching the exploitation of sustainable and ecological energy resources is a key issue [6]. Thermoelectric materials and devices, based on the direct conversion of energy between heat and electricity, were considered as promising alternatives to face the challenges of the global energy dilemma [7], [8]. Thermoelectric materials are capable of converting heat into electricity,

commonly used in cooling and refrigeration applications [9]. A low-cost and scalable thermoelectric material, with a high energy conversion performance, is highly desired [10].

Thermoelectric devices, however, produce tiny voltages and therefore need to be miniaturized and highly integrated with existing semiconductor chip systems with microgenerators or micro refrigerators [11]. However, part of the difficulty in finding thermoelectric materials is their characterization which would allow performance improvements. The efficiency of thermoelectric materials is assessed by the dimensionless merit figure ZT , which is expressed as $ZT = S^2 T / \rho \kappa$. Here, S is the Seebeck coefficient with the unit $[\text{VK}^{-1}]$; ρ is the electrical resistivity with the unit $[\Omega \cdot \text{cm}]$; T is the temperature with the unit $[\text{K}]$; κ is the thermal conductivity with the unit $[\text{W}/(\text{m}^{-1} \cdot \text{K}^{-1})]$. Thus, to obtain the figure of merit of a material, it is necessary to measure S , ρ , and κ .

In the last two decades, several reviews have been produced on thermoelectric materials, including topics such as designing guidances [7], [12]–[14], novel materials [15]–[17] and, in particular, on methods of measurement [1], [18]–[21]. Developments on measurement techniques should be considered as important as the development of new materials. Sometimes, there are large deviations between experimental results and theoretical predictions [1]. For this reason, the importance of a single equipment that calculates all associated thermoelectric quantities allows reducing systematic errors.

This manuscript presents the development of a complete system for the characterization of thermoelectric generation devices and materials. Such a system performs the measurements of electrical resistivity, Seebeck coefficient, and thermal conductivity in a single piece of equipment that included the development of software for control and measurement. The combination of these three quantities gives us information about the Figure of Merit, a thermoelectric quantity of great importance for the tuning of materials thermoelectric performance, i.e., to improve energy generation. To evaluate the here proposed setup, a thin film of indium-tin oxide (ITO) was analyzed. From the measurements and analyses, the equipment proved to be promising for use in industry and scientific research.

II. DETERMINATION OF THE FIGURE OF MERIT

As already mentioned, to obtain the Figure of Merit, it is necessary to measure the electrical resistivity, the Seebeck coefficient, and the thermal conductivity. Here, we will briefly explain how these measurements will be assessed

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The authors are with the Universidade Federal de Itajubá, Itajubá, Minas Gerais 37500-903, Brazil (e-mail: joaoider@gmail.com).

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and their importance in the characterization of thermoelectric generators.

A. Electrical Resistivity

Electrical resistivity is a physical quantity necessary to be determined for the use of materials for electrical and electronic applications, especially if they are films, sheets, and conductive surfaces [22]. The method most extensively used for this purpose is the four-probe system [23], [24] which consists of measuring the current and voltage using independent electrodes from which the resistivity can be obtained according to the sample geometry.

The four-probe system is widely used for the measurement of resistivity and resistances of bulk materials and thin films. This method is not new, having already been used in 1915 to measure the resistance of the Earth [25], known as the Wenner method. However, Valdes [26] adopted the technique to perform electrical resistivity measurements in semiconductor wafers, and, for this reason, it is known today as an analytical method for electrical resistivity measurements in solids, mainly semiconductors. The four-probe method is the most common approach to obtain electrical conductivity [27], [28]. The system consists of a sensor formed by four vertical electrodes, whose tips are coplanarly, collinearly, and regularly spaced placed on the sample surface. The two external electrodes are connected to a current source and the inner ones to a voltmeter [29]. However, it is possible to use other geometric arrangements, as well as to inject current through the internal electrodes [22], [29].

During the measurements, one way to improve the probe contact with the sample surface is the use of springs contacts that provides a certain pressure to the sample surface. Thus, it is possible to always assure contact between the four tips and the sample. This method minimizes measurement errors due to the low contact resistance between the electrodes and the sample [30], [31]. The electrical resistivity measurement is also useful in characterizing the electrical properties of conductive surfaces, such as the change in conductivity due to surface oxidation, as described by Petersen [32].

In general, the voltage V at a function of distance l of an electrode that carries a current i in an electrical resistivity material ρ is given by

$$V = \frac{\rho i}{2\pi f(l)}. \quad (1)$$

Considering a material of semi-infinite dimensions, with the current going in through one end (called 1) and going out through the opposite side (called 4), the voltage V' could be considered as a first attempt of $f(l)$ as

$$V' = \frac{\rho i}{2\pi} \left(\frac{1}{l_{12}} - \frac{1}{l_{34}} \right). \quad (2)$$

Since l_{12} and l_{34} are the distances of the points 1 to 2 and 3 to 4, respectively. By determining the value of potential at the probes V_n for n as the index of probes 2 and 3 ends,

we have to

$$V_2 = \frac{\rho i}{2\pi} \left(\frac{1}{l_{12}} - \frac{1}{l_{23} + l_{34}} \right) \quad (3)$$

$$V_3 = \frac{\rho i}{2\pi} \left(\frac{1}{l_{12} + l_{23}} - \frac{1}{l_{34}} \right). \quad (4)$$

Thus, the difference in total potential, $V = V_2 - V_3$ is equivalent to

$$V = \frac{\rho i}{2\pi} \left(\frac{1}{l_{12}} + \frac{1}{l_{34}} - \frac{1}{l_{23} + l_{34}} - \frac{1}{l_{12} + l_{23}} \right) \quad (5)$$

since what interests us is the resistivity ρ , rearranging (5) we have

$$\rho = \frac{2\pi \frac{V}{i}}{\frac{1}{l_{12}} + \frac{1}{l_{34}} - \frac{1}{l_{23} + l_{34}} - \frac{1}{l_{12} + l_{23}}} \quad (6)$$

in our case the spacing between the probe tips is equal, so $l_{12} = l_{23} = l_{34} = l$ and so (6) is reduced to

$$\rho = 2\pi l \frac{V}{i} \quad (7)$$

this is the general equation. However, there are situations where we do not have a material of semi-infinite dimensions and, therefore, the previous equation (7) must be corrected for a finite geometry.

For an arbitrarily shaped sample, the electrical resistivity is given by (8), F being the correction factor that depends on the shape of the sample, its dimensions, where the tips for the measurement are placed, and also, how it is arranged (type of substrate, for example, in the case of deposited material)

$$\rho = 2\pi l \frac{V}{i} F. \quad (8)$$

F can be derived also for other geometries as carried by Valdes, Schroder, Smits, and Uhler [26], [27], [30], [33].

For our case, the ITO sample is a rectangular thin film, with finite length a , width d , and thickness w , presented in Fig. 1. Being $w < 4l/10$, the F correction factor to be used

$$F = \frac{\pi}{\pi \frac{l}{d} + \ln \left(\frac{1 - e^{-\frac{4\pi l}{d}}}{1 - e^{-\frac{2\pi l}{d}}} \right) + e^{-\frac{2\pi l}{d}} \left(\frac{1}{4} - 2 \right) \frac{(1 - e^{-\frac{6\pi l}{d}})(1 - e^{-\frac{2\pi l}{d}})}{1 + e^{-\frac{2\pi l}{d}}}}. \quad (9)$$

Fig. 1 shows the four-probe in line, three thermocouples, and the dimensions on the rectangular ITO sample. Sample dimensions are w the thickness, a the length, and d the width of the sample. l indicates the distance between consecutive probes. Thermocouples were positioned parallel to the first and last probes and another at the sample center.

B. Seebeck Coefficient

In 1821, the German physicist Thomas Johann Seebeck observed that a compass needle was deflected from its position when it was placed in the vicinity of a closed circuit, where the circuit was simply formed by a link of two different metals and whose junctions were maintained at different temperatures [34]. The temperature difference causes, in the material, the diffusion of charge carriers (electrons or holes) from the hot junction to the cold junction. A configuration

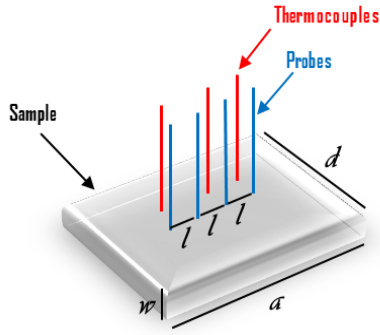


Fig. 1. Illustration of four-probe, three thermocouples placed parallel to the contacts on the extremes and at the center and the dimensions on the rectangular ITO sample. All sample dimensions and contact distances are represented in terms of the equation parameters.

of mobile charge carriers migrate to the cold side and leave behind the immobile cores of opposite charge on the hot side, thus giving rise to a voltage difference called thermoelectric potential. The value of the thermoelectric electromotive force f_{em} , which is generated between the two different materials, such as metals or semiconductors, can be obtained

$$E_{AB} = S \Delta T \quad (10)$$

where S is defined as the relative Seebeck coefficient between materials A and B. The (10) can be rewritten according to the potential difference generated, knowing that $E_{AB} = -\Delta V$

$$S = -\frac{\Delta V}{\Delta T}. \quad (11)$$

This leads to a direct current that flows through the materials, through the junction, when they are kept at different temperatures. The electromotive force E_{AB} and the coefficient S are positive if the conventional current flows from the material A to B from the hot junction.

Two approaches are commonly used to measure the Seebeck coefficient in the thin-film plane: the integral method and the differential method. In the integral method, [35] one end of the sample is kept at a reference temperature (usually room temperature or 273 K) and the other is increased continuously over a temperature range of interest. This method assumes that the thermal balance between the thermocouple junctions and the sample is achieved instantly, and there is no delay between the voltage and temperature measurements.

The differential method, used in this work, for measuring the Seebeck coefficient in the plane is more widely used for thin-film thermoelectric materials [36]–[38]. Small thermal gradients are applied along the length of the sample, close to the temperature of interest; that is, in the linear regime. Several data points with variable temperature differences are collected and the slope of the voltage about the temperature difference provides the Seebeck coefficient.

C. Thermal Conductivity

The measurement of thermal conductivity is performed using the method created by Kohlrausch [39]. It is a

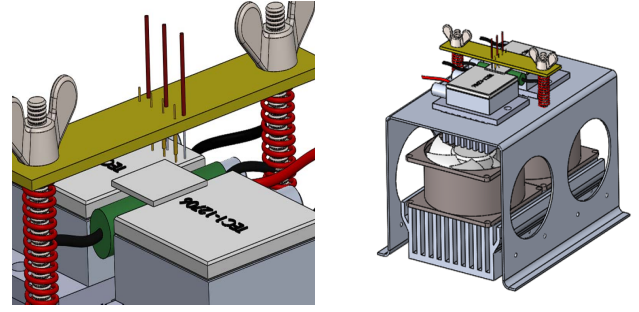


Fig. 2. Illustration of the thermoelectric sample characterization system. In the image it is possible to check the circular air vents on the side, increasing the efficiency in cooling the system.

well-established method in the physical and thermodynamic analysis of materials, presented in [40] and [41].

The Kohlrausch method shows how constant temperature can be used to find the ratio of thermal (κ) and electrical (σ), known as the Wiedemann–Franz ratio [42]. The extreme regions of the sample are kept as close as possible to the same temperature by making $T_1 = T_2 = T$, while the center of the sample is heated to T_h , thus generating a temperature gradient $\Delta T = T_h - T$. This generates a potential difference ΔV between the ends of the sample and (12) describes the entire phenomenon

$$\frac{\kappa}{\sigma} = \frac{(\Delta V)^2}{8 \Delta T} \quad (12)$$

with this, it is possible to determine simply the κ/σ ratio of the thermal and electrical conductivities, involving only the reading of the temperature difference and the potentials in two points of the sample.

D. Figure of Merit

With the measurements of electrical conductivity, Seebeck coefficient, and thermal conductivity it is possible to obtain the Figure of Merit ZT

$$ZT = \frac{S^2 T}{\rho \kappa} \quad (13)$$

where S is the Seebeck coefficient, ρ the electrical resistivity, κ the thermal conductivity and T the temperature, which leaves the ZT dimensionless.

III. EXPERIMENTAL SETUP

The developed system has a structure assembled to carry out electrical conductivity measurements, Seebeck coefficient, and thermal conductivity in the same equipment. This excludes errors due to the use of various equipment and increases the robustness of the measurements.

In the system we developed, heatsinks, fans, water blocks, Peltier devices, a Teflon block, a power resistor, and a metallic structure that provides stability and support for the entire system were used. All characterization system is done as illustrated in detail in the Fig. 2, alternating only with the rectangular Teflon block, to perform the electrical conductivity

measurements and the Seebeck coefficient, and the power resistor, to perform the thermal conductivity measurement.

To carry out the Seebeck coefficient and thermal conductivity measurements, temperature differences must occur. The water blocks, where cold water is circulated, have the function of maintaining a stable temperature in the Peltier modules, generating a temperature gradient and preventing it from entering thermal equilibrium. Thermal contact between the Peltier modules and the sample is mechanically assured by a couple of coils besides the extra contact pin coils.

The heatsinks under the Peltier modules function to dissipate excess heat from the aluminum water blocks. The two fans under the smaller heatsinks under aluminum waterblocks are used to force an airflow from top to bottom, assisting the heatsink that is at the bottom of the structure. In this way, the heat generated by the Peltier modules is efficiently dissipated out of the system.

As illustrated in Fig. 2, the structure has two circular air vents on its sides, the front and rear part is open, to allow air to circulate efficiently.

A. Electrical Conductivity and Seebeck Coefficient

To perform the electrical conductivity measurement, the four-probe technique was used, where four spring contact pins model 0985-2-15-20-71-14-11-0 brand Mil-max 0.48 mm and limited to 2-A current, in diameter and in line are placed in contact on the sample surface (see Fig. 1). The pins were equally spaced by 3.9 mm so that minimal sample length $a \geq 14$ mm and the width $d \geq 4$ mm in order to accommodate pins and thermocouples. The sample is on a surface, formed by the two Peltier devices and the Teflon block. The Teflon block is necessary to give mechanical support to the sample between the Peltier modules and also to avoid them to enter into thermal equilibrium. An external voltage is applied through the two external pins, using the 5 1/2 digit Keithley 2400 source-measure unit, from which the current is also measured. The minimal applied current is about $1 \mu\text{A}$ and maximal is 1 A but limited to 20 V. We measure the voltage using the two internal pins, using the 6 1/2 digit Keithley 2100 multimeter with its resolution limited to $0.1 \mu\text{V}$ at 0.1-V range and 1 mV at 1000-V range, as illustrated in Fig. 3. With the $I(V)$ curves of the material, we obtain, according to (8), the electrical resistivity of the sample.

The measurement of the Seebeck coefficient is carried using two Peltier modules, placed under the sample. One module is the hot temperature reference (limited to 390 K) and the other is the cold temperature reference (limited to 280 K). The purpose here is to generate a temperature gradient in the sample. Voltage is applied to Peltier devices through the Control Software, which was developed in Python. The software controls an Arduino Mega 2560, which sends a voltage to a low-pass filter. This low-pass filter converts the received signal to two independent Bk Precision XLN3640 sources, each of which will apply a voltage to a Peltier device. The water blocks are connected by hoses to a cooling system, formed by Cold Water Reservoir, the Digital Temperature Controller,

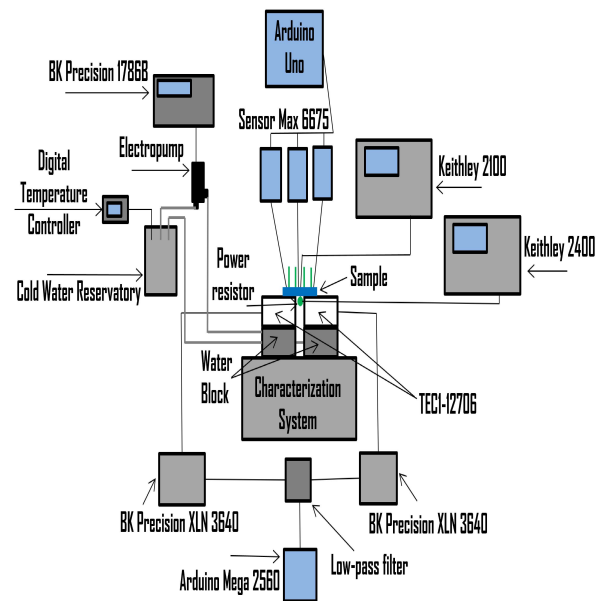


Fig. 3. Scheme for measuring electrical resistivity, Seebeck coefficient, and thermal conductivity.

the Electropump, and the BK Precision 1786B that provides the necessary voltage.

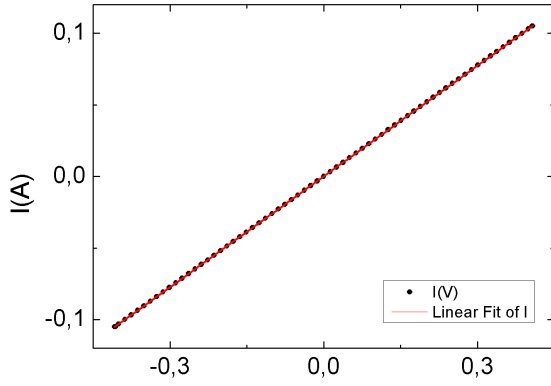
The procedure is then to generate a temperature gradient in the material so that by the Seebeck effect the generation of voltage occurs. The voltage generated is measured by Keithley 2100, through the two external pins connected to the sample. The Arduino Uno, through the Max6675 Sensors, performs the reading of the temperatures in the sample, to analyze the data according to (11). Fig. 3 shows the assembly diagram for measurements of the Seebeck coefficient.

B. Thermal Conductivity

The measurement of thermal conductivity is carried out by the Kohlrausch method that uses two Peltier modules being at the same cold reference temperature, and a power resistor in place of the Teflon block in the center of the sample to be the hot temperature reference.

The power resistor is activated, by Keithley 2400, to raise the temperature of the sample center, while the Peltiers cool the external parts, from the Arduino Mega2560, the low-pass filter, and the two BK Precision XLN3640. Peltier devices are also cooled by the cooling system, composed of Cold Water Reservoir, the Digital Temperature Controller, the Electropump, and the BK Precision 1786B.

The voltage generated between the outer pins on the surface of the material is measured with the Keithley 2100 multimeter. The temperatures on the material surface points are very important to be taken close to the measuring pins using three K type thermocouples and an Arduino Uno connecting each thermocouple to a Max6675 transducer with 0.25-K temperature resolution. Typical tip ball diameter is of the order of 0.5 mm, the same as the spring electrical pins. One thermocouple is placed close to the center pin and the other

Fig. 4. $I(V)$ curve obtained for calculating the electrical resistivity ρ , of ITO.

two close to the outer ones. With the data collected, simply apply (3) and obtain the thermal conductivity of the material.

For an analysis of the operation of the equipment, a sample of Indium tin oxide, commonly referred to as ITO, was used which is a standard degenerate n-type transparent semiconductor material. It is highly transparent to visible light ($>80\%$) and low resistivity ($\rho < 5 \times 10^{-4} \Omega \cdot \text{cm}$) [43], [44], being $\rho \approx 1 \times 10^{-4} \Omega \cdot \text{cm}$ in [45], and ranged from $2.9 \times 10^{-4} \Omega \cdot \text{cm}$ to $3.2 \times 10^{-3} \Omega \cdot \text{cm}$ in [46].

Its Seebeck coefficient varies from 5.96 to $43.16 \mu\text{VK}^{-1}$, depending on the material manufacturing process [47]. Thermal conductivity ranged to 4.00 to $5.86 \text{ W}/(\text{m}^{-1} \cdot \text{K}^{-1})$ according to material growth [46]. In [48] we have $\kappa \approx 0.84 \pm 0.12 \text{ W}/(\text{m}^{-1} \cdot \text{K}^{-1})$ at room temperature for nanogranular ITO, can be $\kappa \approx 10 \text{ W}/(\text{m}^{-1} \cdot \text{K}^{-1})$ at room temperature [49], [50], $\kappa \approx 1.2 \text{ W}/(\text{m}^{-1} \cdot \text{K}^{-1})$ at 973 K in sintered In_2O_3 doped by Zn and Ce [50]. In [51] and [52] $\kappa \approx 1.58 - 1.75 \text{ W}/(\text{m}^{-1} \cdot \text{K}^{-1})$.

The Z-factor varies according to the thickness of the material, being for 50, 100, 200, 280 nm, respectively, 9.76×10^{-5} , 7.16×10^{-4} , 1.28×10^{-3} , and $1.84 \times 10^{-3} \text{ K}^{-1}$ [53]. In [54] the Z-factor is $1.19 \times 10^{-2} \text{ K}^{-1}$.

Our sample is a rectangular thin ITO film deposited onto a glass substrate $w = 10\text{-nm}$ thick. The spacing between the four points of the measurement is $l = 3.9 \text{ mm}$, the length of the rectangular sample (in the direction of the four points) is $a = 24.9 \text{ mm}$ and the width of the rectangular sample (perpendicular to the four points) is $d = 17.1 \text{ mm}$.

IV. RESULTS

To evaluate the efficiency of the equipment, a sequence of tests was performed on a thin ITO film, which is already well established in the literature. We performed the measurements of electrical conductivity, Seebeck coefficient, and thermal conductivity, to obtain the Figure of Merit.

Five $I(V)$ curves were made for the sample. Fig. 4 shows one of the $I(V)$ curves obtained for calculating the electrical resistivity ρ of ITO.

From the graph, we obtained the resistance $R = V/I$ of the sample, and from (8) we calculated the resistivity of the sample.

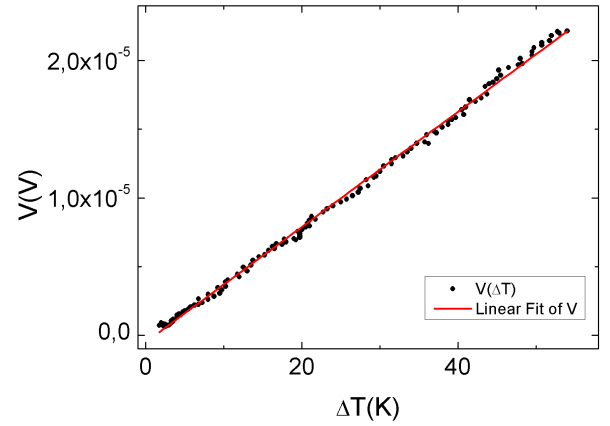
Fig. 5. $V(\Delta T)$ curve obtained to calculate the Seebeck coefficient. $T_C = 293 \text{ K}$.

TABLE I
VALUES MEAN OF THE ELECTRICAL RESISTIVITY, SEEBECK
COEFFICIENT, AND THERMAL CONDUCTIVITY
OF THE ITO SAMPLE

Parameter	Literature	Experimental
$\rho (\Omega\text{cm})$	1×10^{-4} [45] $\rightarrow 3.2 \times 10^{-3}$ [46]	$(1.3100 \pm 0.0007) \times 10^{-4}$
$S (\mu\text{VK}^{-1})$	$5.96 \rightarrow 43.16$ [47]	(4.38 ± 0.22)
$\kappa (\text{Wm}^{-1}\text{K}^{-1})$	0.84 [48] $\rightarrow 10$ [50]	(3.10 ± 0.83)
$Z (\text{K}^{-1})$	$2.98 \times 10^{-5} \rightarrow 5.82 \times 10^{-4}$	$(4.7 \pm 1.4) \times 10^{-5}$

With a statistical calculation, we obtain the electrical resistivity of the material, presented in Table I. Five measurements were also taken to calculate the Seebeck S coefficient. Fig. 5 shows one of the five $V(\Delta T)$ curves obtained.

From statistical calculations, the value of the Seebeck S coefficient of the ITO sample was obtained, presented in Table I. The values of thermal conductivity were calculated from five measurements and (12) are shown in Table I. Thus, Table I presents all the experimental data obtained from our analysis and those presented in the literature. The value of Z was calculated by (13) using the values of reference [46]–[48], [50] and presented in the same Table. As can be seen in the comparison of the values obtained in the literature and the results obtained by our equipment, the equipment presented an excellent performance and obtained outstanding results, showing its robustness for applications in the characterization of thermoelectric devices.

V. CONCLUSION

The single equipment shown here reduces the time and complexity considered in the measurement of all the three thermoelectric quantities necessary to obtain the figure of merit since, for that task, it is commonly used at least two distinct equipment one for the Seebeck coefficient and electrical resistivity and another for the thermal conductivity. The presented equipment was designed to perform accurate temperature control leading to stable measurements. The values of electrical resistivity, Seebeck coefficient, and thermal conductivity obtained by our robust thermoelectric characterization system will contribute to improving access to the determination of the figure of merit in a simple, fast and reliable form, which will contribute to research on the characterization of thermoelectric materials.

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Adhimar Oliveira received the bachelor's degree in physics from the Universidade Federal de São João del-Rei (UFSJ), São João del-Rei, Brazil, and the master's degree in physics and applied mathematics in astrophysics and the Ph.D. degree in materials science for engineering from the Universidade Federal de Itajubá (UNIFEI), Itajubá, Brazil.

He is currently a Professor, a Coordinator of the physics degree course at a distance, and a Professor of the postgraduate program in physics with UNIFEI.



João Ider received the bachelor's degree and the master's degree in physics from the Federal University of Itajubá (UNIFEI), Itajubá, Brazil, where he is currently pursuing the Ph.D. degree in materials science for engineering.

His research interests include general relativity, condensed matter, and analogous models. It also works with scientific and literary dissemination through its social network, Instagram.



Rero Marques Rubinger is currently a Full Professor with the Universidade Federal de Itajubá, Itajubá, Brazil, where he has been working since 2002 in undergraduate courses of physics for physical sciences and engineering and graduate program of materials science and engineering. He has published about 58 articles in journals. He works with semiconductor physics and devices, focusing on renewable energy materials and devices, such as solar cells.